

SOC Investigation Report — Attempted LSASS Credential Dumping Detection and Investigation

Field	Value
Case ID	SOC-LAB-2026-0727-LSASS
Classification	Detection Engineering Lab / SOC L1 Investigation
Analyst	Edwin Dominic
Host	WIN11-CLIENT01
User Account	WIN11-CLIENT01\Socadmin
Data Source	Sysmon (Operational log) → Splunk (index=sysmon)
Date of Activity	2026-07-27 (18:39:15 – 18:39:27 UTC)
Alert Triggered	<i>Suspicious LOLBin Activity – Rundll32 / comsvcs.dll MiniDump</i>
Severity (Investigated)	High — confirmed credential dumping attempt
Outcome	Attempt did not complete — LSASS was never accessed

1. Objective

Detect and investigate a suspected LSASS memory dump attempt using `rundll32.exe` and `comsvcs.dll MiniDump` — a well-known LOLBin technique (MITRE T1003.001) used to extract credential material from the Local Security Authority Subsystem Service without introducing third-party tooling. The goal was to determine whether the attempt succeeded, reconstruct the execution chain using Sysmon telemetry in Splunk, identify why expected artifacts were absent, and document the findings accurately.

2. Initial Detection

Query: `index=sysmon EventCode=1 Image="*\rundll32.exe" CommandLine="*MiniDump"`

Finding

`rundll32.exe` (PID 2804, ProcessGuid {0875a023-a5df-6a67-5001-000000003c00}) was spawned by `cmd.exe` (PID 9240) at 18:39:27.184 UTC under account `WIN11-CLIENT01\Socadmin`, with the command line:

```
rundll32.exe C:\Windows\System32\comsvcs.dll, MiniDump 956
C:\Windows\Temp\lsass.dmp full
```

All subsequent evidence was correlated using ProcessGuid rather than PID.

3. Initial Assessment — Why the Activity Was Flagged

Indicator	Detail
rundll32.exe invoking comsvcs.dll	comsvcs.dll is a Windows COM+ component — it has no legitimate reason to be called interactively from a command prompt
MiniDump export function	Explicitly exports a process memory dump; legitimate administrative use never requires this from a user command line
Target path C:\Windows\Temp\lsass.dmp	Output written to a world-writable system temp path under an .dmp extension — consistent with credential staging
full argument	Requests a full memory dump, not a heap or mini-dump — maximizes credential material captured
Execution at High integrity	Elevated administrative context required to open an LSASS handle — confirms deliberate, privilege-aware execution
Parent is interactive cmd.exe	Manual, operator-driven execution rather than a scripted or automated background process

4. Parent Process Investigation

Query: index=sysmon EventCode=1 ProcessGuid="{0875a023-a5d3-6a67-4e01-000000003c00}"

Finding

cmd.exe (PID 9240, ProcessGuid {0875a023-a5d3-6a67-4e01-000000003c00}) was launched by explorer.exe (PID 5704) at 18:39:15.952 UTC, confirming an interactive, user-initiated command prompt — not a service, scheduled task, or lateral-movement delivery mechanism.

Field	cmd.exe	rundll32.exe
PID	9240	2804
ProcessGuid	{0875a023-a5d3-6a67-4e01-000000003c00}	{0875a023-a5df-6a67-5001-000000003c00}
Parent	explorer.exe (5704)	cmd.exe (9240)
Integrity Level	High	High
LogonId	0x7E6AD	0x7E6AD
SHA256	65C9EC61A5B10E7EEC76A7A961766B86BB93604EF18C5D1FE8AC53B4E53EF17A	2370D904F3C18A68638B3AAEDF17B2C68306E98A784EB1F21985C4E880F1D2D

Both processes share the same LogonId (0x7E6AD), confirming they ran under a single interactive logon session.

5. Expected Telemetry — Not Observed

Query: `index=sysmon EventCode=10 TargetImage="*\\lsass.exe"`

Finding

No Sysmon Event ID 10 (Process Access) targeting `lsass.exe` was generated. The Sysmon configuration on this host has `ProcessAccess` set to `onmatch: include` with no sub-conditions — this is a catch-all rule meaning **every** process access event is logged regardless of target. The absence of Event ID 10 is therefore definitively significant: `rundll32.exe` never obtained a handle to the LSASS process.

Query: `index=sysmon EventCode=11 TargetFilename="*lsass.dmp"`

Finding

No Sysmon Event ID 11 (File Create) for `lsass.dmp` was generated. The active Sysmon `FileCreate` rule explicitly includes `.dmp` files (`TargetFilename` end with `.dmp`), meaning any dump file written to disk would have produced a log entry. No such event exists. The file `C:\Windows\Temp\lsass.dmp` was confirmed absent from disk.

6. Root Cause Analysis — Why the Attempt Failed

Investigation of the registry output collected from the host identified two contributing factors.

Primary Root Cause — Wrong LSASS PID Targeted

The command specified PID `956` as the LSASS target:

```
MiniDump 956 C:\Windows\Temp\lsass.dmp full
```

The registry key `HKLM\SYSTEM\CurrentControlSet\Control\Lsa\LsaPid` returned:

```
LsaPid      REG_DWORD    0x3C8
```

`0x3C8` decimal = **968**. The actual running LSASS PID at the time was **968**, not 956. The `MiniDump` call targeted a process that was either a different process entirely or non-existent, so it never opened an LSASS handle. This is why no Event ID 10 was generated and why no dump file was produced.

Secondary Defense — UEFI-Enforced PPL

Even if the correct PID had been used, the registry also showed:

```
RunAsPPLBoot  REG_DWORD    0x2
IsPplAutoEnabled  REG_DWORD    0x1
```

`RunAsPPLBoot = 2` indicates LSASS Protected Process Light (PPL) is enforced at the UEFI/firmware boot level — this overrides the `RunAsPPL = 0` software registry value and cannot be disabled from a running OS session. An unsigned tool like a plain `rundll32.exe / comsvcs.dll` call cannot open a handle to a PPL-protected process at all. Had the correct PID been supplied, this protection layer would have blocked the access attempt independently.

7. Attack Chain

```
explorer.exe (PID 5704)
├─ cmd.exe (PID 9240) – interactive, High integrity
│   └─ rundll32.exe (PID 2804) – comsvcs.dll MiniDump 956
│       └─ [target PID 956 ≠ LSASS PID 968 – no LSASS handle obtained]
│           [no Event ID 10]
│           [no Event ID 11]
│           [no lsass.dmp on disk]
```

8. Timeline Reconstruction (UTC)

Time	Activity
18:39:15.952	explorer.exe spawns cmd.exe (PID 9240) at High integrity under Socadmin
18:39:27.184	cmd.exe spawns rundll32.exe (PID 2804) with comsvcs.dll MiniDump 956 C:\Windows\Temp\lsass.dmp full
18:39:27 (observed)	No Sysmon Event ID 10 generated — rundll32.exe did not open LSASS
18:39:27 (observed)	No Sysmon Event ID 11 generated — lsass.dmp was not written to disk

Total elapsed time from cmd.exe launch to dump attempt: ~12 seconds (consistent with manual, interactive execution).

9. MITRE ATT&CK Mapping

Tactic	Technique	ID	Evidence
Credential Access	OS Credential Dumping: LSASS Memory	T1003.001	rundll32.exe invoked comsvcs.dll MiniDump targeting an LSASS-intended PID
Execution	System Binary Proxy Execution: Rundll32	T1218.011	A trusted, signed Windows binary was used to invoke the dump function rather than external tooling

10. Indicators of Compromise (IOC)

Process Indicators

Process	PID	Parent	SHA256
cmd.exe	9240	explorer.exe (5704)	65C9EC61A5B10E7EEC76A7A961766B86BB93604EF18C5D1FE8AC53B4E53EF17A
rundll32.exe	2804	cmd.exe (9240)	2370D904F3C18A68638B3AAEDF17B2C68306E98A784EB1F21985C4E880FF1D2D

Command-Line Indicators

Artifact	Value
Suspicious DLL	C:\Windows\System32\comsvcs.dll
Invoked export	MiniDump
Intended dump path	C:\Windows\Temp\lsass.dmp
Dump type	full

11. SOC Verdict

Detection: Successful — suspicious activity was identified from Sysmon Event ID 1 alone.

Execution: Successful — rundll32.exe with comsvcs.dll MiniDump ran to completion.

Credential Access: Not confirmed — LSASS was never accessed. No Event ID 10, no dump file.

Impact: No evidence of credential exposure from this attempt.

12. Splunk Threat Hunting & Correlation Queries

Detect comsvcs.dll MiniDump Attempts (scheduled alert)

```
index=sysmon EventCode=1 Image="*\\rundll32.exe" CommandLine="*comsvcs*"
CommandLine="*MiniDump*"
| table _time Computer User ParentImage CommandLine IntegrityLevel ProcessGuid
```

Purpose: High-fidelity alert with near-zero false-positive rate — legitimate business activity never calls comsvcs.dll MiniDump from a command line.

Confirm Whether LSASS Was Accessed

```
index=sysmon EventCode=10 TargetImage="*\\lsass.exe"
| table _time Computer User SourceImage TargetImage GrantedAccess CallTrace
```

Purpose: Determines whether the attempt progressed past process creation to actually opening an LSASS handle — the critical pivot point between an attempt and a success.

Confirm Whether a Dump File Was Created

```
index=sysmon EventCode=11 TargetFilename="*.dmp"
| table _time Computer User Image TargetFilename
```

Purpose: Closes the investigation by confirming or ruling out dump file creation on disk.

Correlate the Full Process Lineage

```
index=sysmon (ProcessGuid="{0875a023-a5d3-6a67-4e01-0000000003c00}" OR
ParentProcessGuid="{0875a023-a5d3-6a67-4e01-0000000003c00}")
| sort _time
```

```
| table _time EventCode Image ParentImage CommandLine IntegrityLevel
```

Purpose: Reconstructs the full chain from `explorer.exe` through `cmd.exe` and `rundll32.exe` in one pass.

Broad Hunt for LOLBin-Based LSASS Targeting (environment-wide)

```
index=sysmon EventCode=1 (CommandLine="*lsass*" OR CommandLine="*MiniDump*" OR
CommandLine="*comsvcs*")
| stats count by Computer User ParentImage CommandLine
```

Purpose: Scans for any variation of this technique across the environment, not just this specific execution.

13. Detection Engineering

Monitoring coverage — this activity is reliably detectable by alerting on:

- `rundll32.exe` command lines containing `comsvcs.dll` and `MiniDump` together
- Any process opening `lsass.exe` with a read-memory access mask (Event ID 10, `GrantedAccess` values such as `0x1010`, `0x1410`, `0x1fffff`)
- `.dmp` files written to user-accessible paths (`%TEMP%`, `C:\Windows\Temp`, user profile directories)
- `rundll32.exe` or `cmd.exe` invocations where the command line contains an LSASS-associated PID

Actionable rule specs:

1. **High-fidelity alert:** any `rundll32.exe` execution with `comsvcs` and `MiniDump` in the same command line — alert immediately, no additional correlation needed before escalating.
2. **Process access rule:** alert on Event ID 10 where `TargetImage` ends with `lsass.exe` and `GrantedAccess` includes read-memory permissions — this fires only if the access actually reaches LSASS and is the definitive success indicator.
3. **Chained rule:** correlate Event ID 1 (`MiniDump` command) → Event ID 10 (LSASS handle) → Event ID 11 (`.dmp` file created) within a 60-second window to classify confirmed success vs. blocked attempt automatically.

Detection Engineering Queries:

Detect Any LSASS Process Access (Event ID 10)

```
index=sysmon EventCode=10 TargetImage="*\\lsass.exe"
| stats count by Computer SourceImage GrantedAccess CallTrace
```

Detect Dump Files Written to Sensitive Paths

```
index=sysmon EventCode=11 TargetFilename="*.dmp" (TargetFilename="*Temp*" OR
TargetFilename="*Users*" OR TargetFilename="*Public*")
| table _time Computer Image TargetFilename
```

14. Recommended Response Actions

If this pattern were confirmed in a production environment:

1. **Escalate immediately** on the Event ID 1 alone — this command line pattern has no legitimate use case and should not wait for Event ID 10 or dump file confirmation before being treated as high severity.
 2. **Check for the dump file** at the specified path and any alternative paths (`%TEMP%`, user profile, USB-mounted paths).
 3. **Query for Event ID 10** targeting `lsass.exe` within the same session window — absence does not close the case if other dump tools were subsequently used.
 4. **Review the full session** for follow-on activity: exfiltration of any dump file, lateral movement tools (`psexec`, `wmiexec`), or privilege escalation indicators.
 5. **Validate the active LSASS PID** (`LsaPid` registry key) against the PID used in the command — a mismatch, as observed here, is itself an indicator of an inexperienced or scripted attacker who did not confirm the target PID before executing.
 6. **Reset credentials** for all accounts whose sessions were active on the host, regardless of whether the dump was confirmed — the intent was credential access.
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15. Conclusion

A credential dumping attempt targeting LSASS was detected via Sysmon Event ID 1 showing `rundll32.exe` invoking `comsvcs.dll MiniDump` from an interactive, High-integrity administrative session. Investigation confirmed that the attempt did not complete — no Sysmon Event ID 10 (Process Access) was generated, and no dump file was created.

Root cause analysis identified two reasons: the command targeted PID 956 rather than the actual running LSASS PID 968 (confirmed via `LsaPid REG_DWORD 0x3C8` from the host registry), meaning `MiniDump` never opened an LSASS handle; and independently, UEFI-enforced PPL (`RunAsPPLBoot = 0x2`) would have blocked an unsigned caller from accessing LSASS even with the correct PID.

This investigation demonstrates that a detection can be successful on a single Event ID 1 alone, that the absence of expected follow-on telemetry (Event ID 10, Event ID 11) is itself an analytical finding rather than a gap, and that root-cause analysis — rather than simply noting "the attack failed" — produces the actionable intelligence that separates a complete investigation from an incomplete one.